

[65811]-1909
F.E. (Common)
ENGINEERING MATHEMATICS - II
(2019 Course) (Semester - II) (107008)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Q.1 is compulsory.
- 2) Solve Q.2 or Q.3, Q.4 or Q.5, Q.6 or Q.7, Q.8 or Q.9.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Figures to the right indicate full marks.
- 5) Use of electronic pocket calculator is allowed.
- 6) Assume suitable data, if necessary.

Q1) Write the correct options for the following multiple choice questions.

a) $\int_0^{\pi/2} \cos^4 x \, dx =$

[2]

i) 0

ii) $\frac{3}{8}\pi$

iii) $\frac{3}{16}\pi$

iv) $\frac{\pi}{2}$

b) The curve $y^2 = x^2(2 - x)$ is symmetric about

[2]

i) X-axis

ii) Y-axis

iii) both X & Y axis

iv) Origin

c) Centre (C) and radius (r) of the sphere $x^2 + y^2 + z^2 - 2x - 2y - 2z - 6 = 0$ are

[2]

i) $C \equiv (1, 1, 1), r = \sqrt{3}$

ii) $C \equiv (-1, -1, -1), r = 3$

iii) $C \equiv (1, 1, 1), r = 3$

iv) $C \equiv (-1, -1, -1), r = 9$

d) $\int_{y=0}^{y=1} \int_{x=0}^2 x \, dx \, dy = \dots$

[2]

i) 0

ii) 1

iii) 2

iv) 4

- e) Total number of loops for the curve $r = a \sin 3\theta$ are [1]
- i) 2 ii) 3
iii) 4 iv) 5

Q2) a) If $U_n = \int_0^{\pi/4} \tan^n \theta d\theta$, then Prove that, $n(U_{n+1} + U_{n-1}) = 1$ **[5]**

c) Evaluate, $\frac{d}{dx} \operatorname{erf}_c(\sqrt{x})$ [5]

Q3) a) If $I_n = \int_0^{\pi/3} \cos^n x dx$ then prove that,

$$I_n = \frac{\sqrt{3}}{n \cdot 2^n} + \frac{n-1}{n} I_{n-2}$$

[5]

c) Prove that $\int_0^1 \frac{x^a - 1}{\log x} dx = \log(a+1)$ [5]

S being measured from the cusp which lies on Y axis

OR

- Q5)** a) Trace the curve $x^2y^2 = a^2(y^2 - x^2)$ [5]
b) Trace the curve $\gamma = a\cos(3\theta)$ [5]

- c) Trace the curve $x = t^2, y = t - \frac{t^3}{3}$ [5]

- Q6)** a) Show that the plane $2x - 2y + z + 12 = 0$ touches the sphere $x^2 + y^2 + z^2 - 2x - 4y + 2z - 3 = 0$ and find the point of contact. [5]

- b) Find the equation of right circular cone whose vertex is at $(1, -1, 2)$, axis is the line $\frac{x-1}{2} = \frac{y+1}{1} = \frac{z-2}{-2}$ and semivertical angle 45° . [5]

- c) Find the equation of right circular cylinder whose axis is $\frac{x}{2} = \frac{y}{1} = \frac{z}{-2}$ and radius is 4. [5]

OR

- Q7)** a) Find the sphere through the circle, $x^2 + y^2 + z^2 = 4, z = 0$, meeting the plane $x + 2y + 2z = 0$ in a circle of radius 3. [5]

- b) Find the equation of the right circular cone whose vertex is at origin, axis is y axis and semivertical angle 30° . [5]

- c) Find the equation of right circular cylinder of radius 2, whose axis passes through origin and makes equal angles with coordinate axes. [5]

- Q8)** a) Evaluate $\int_0^1 \int_{x^2}^x xy(x+y) dx dy$ [5]

- b) Find the area bounded by the curves $y^2 = 4ax$ & $x^2 = 4ay$ [5]

- c) A thin plate of uniform thickness and density is in the form of lamina bounded by the parabola $x^2 = y$ and the line $y = x + 2$. Find the moment of inertia of the lamina about Y - axis. [5]

OR

- Q9)** a) Evaluate $\int_0^1 dx \int_0^2 dy \int_1^2 x^2 yz dz$ [5]

- b) Find the volume of the region enclosed by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$ [5]

- c) Find the centre of gravity of loop of the rose curve $r = a \sin 2\theta$ in the first quadrant. [5]

